

NeuronSeqSampler: A Hybrid Neural-Rhythmic Audio Sequencer

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Abstract

We present NeuronSeqSampler, a novel real-time audio sequencing system that combines artificial neural network principles with rhythm interpretation for generative music creation. The system employs a user-configurable spiking neural network coupled with a multi-band rhythmogram analyzer to create adaptive, rhythm-driven sample triggering. Drawing on principles from Todd’s (1994) auditory primal sketch for rhythmic grouping and Laine’s (2000) neural modeling approach for musical note analysis, our implementation features adaptive learning algorithms, onset detection, and tempo-relative frequency analysis. We describe the mathematical foundations, signal processing architecture, and learning mechanisms that enable the system to respond to and generate complex rhythmic patterns.

1 Introduction

Generative music systems have traditionally operated either through rule-based algorithmic composition or through neural network approaches trained on large datasets. NeuronSeqSampler bridges these paradigms by implementing a real-time system where a configurable neural network responds to rhythmic features extracted from audio input, creating an interactive performance tool that learns and adapts to musical context.

The system architecture consists of three primary components:

1. A rhythmogram analyzer implementing multi-scale temporal decomposition [2]
2. A recurrent spiking neural network with configurable topology
3. An adaptive learning system with onset-weighted updates

2 Rhythmogram Analysis

2.1 Theoretical Foundation

Following Todd’s (1994) auditory primal sketch model [2], we implement a multi-scale rhythmic analysis that decomposes the input signal into musically relevant frequency bands. Unlike traditional spectral analysis, the rhythmogram focuses on temporal periodicities corresponding to musical rhythms.

2.2 Filter Bank Architecture

The system employs a logarithmically-spaced bank of bandpass filters covering the range 0.125 Hz to 16 Hz, corresponding to musical tempos from approximately 7.5 to 960 BPM:

$$f_i = f_{\min} \cdot \left(\frac{f_{\max}}{f_{\min}} \right)^{i/(N-1)}, \quad i = 0, 1, \dots, N-1 \quad (1)$$

where N is the number of bands (default 8), $f_{\min} = 0.125$ Hz, and $f_{\max} = 16$ Hz.

Each band i is implemented as a second-order IIR bandpass filter with transfer function:

$$H_i(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}} \quad (2)$$

The bandwidth of each filter is set proportionally to its center frequency:

$$BW_i = k \cdot f_i \quad (3)$$

where k is a scaling factor (default 0.5), ensuring constant-Q behavior.

2.3 Onset Detection

For each frequency band, we implement onset detection. The onset detection function computes the spectral flux in each band:

$$\Phi_i(n) = \max(0, |X_i(n)| - |X_i(n-1)|) \quad (4)$$

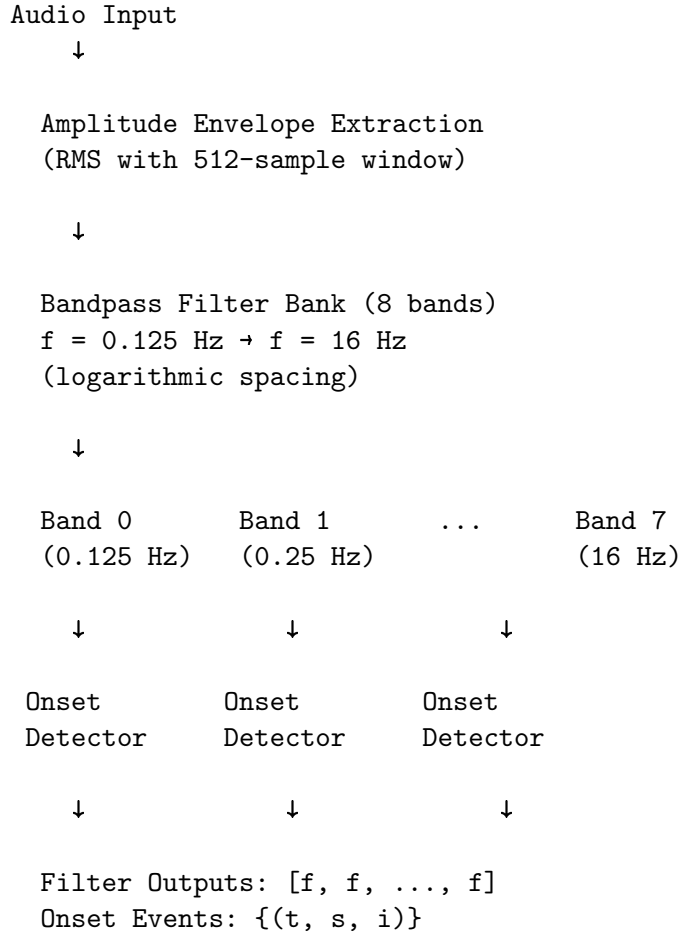
where $X_i(n)$ is the filter output for band i at time n . An onset is detected when:

$$\Phi_i(n) > \theta \cdot \mu_i + \epsilon \quad (5)$$

where θ is a threshold parameter, μ_i is a local mean of the onset function, and ϵ is a small constant to prevent spurious detections during silence.

2.4 Signal Flow Diagram — Rhythmogram

The rhythmogram processing pipeline follows this architecture:



where t=time, s=strength, i=band

3 Neural Network Architecture

3.1 Neuron Model

Each neuron j in the network is modeled as a leaky integrate-and-fire unit following Laine's (2000) [1] neural approach to musical note modeling, with activation function ϕ_j :

$$a_j(t+1) = \phi_j \left(\lambda_j \cdot a_j(t) + \sum_{k \in \mathcal{P}(j)} w_{kj} \cdot s_k(t) + r_j(t) \right) \quad (6)$$

where:

- $a_j(t)$ is the activation of neuron j at time t
- $\lambda_j \in [0, 1]$ is the leak rate (decay factor)
- $\mathcal{P}(j)$ is the set of neurons with connections to j
- w_{kj} is the weight from neuron k to neuron j
- $s_k(t)$ is the spike output of neuron k (1 if fired, 0 otherwise)
- $r_j(t)$ is the rhythmic input to neuron j

The activation function ϕ_j can be selected from:

- Linear: $\phi(x) = x$
- Sigmoid: $\phi(x) = \frac{1}{1+e^{-x}}$
- ReLU: $\phi(x) = \max(0, x)$
- Tanh: $\phi(x) = \tanh(x)$

3.2 Firing Threshold

A neuron fires when its activation exceeds its threshold θ_j :

$$s_j(t) = \begin{cases} 1 & \text{if } a_j(t) \geq \theta_j \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Upon firing, the neuron triggers its associated audio sample and its activation is reset:

$$a_j(t) \leftarrow 0 \quad (8)$$

3.3 Rhythmic Input Mapping

The rhythmic input $r_j(t)$ for neuron j is computed from the rhythmogram outputs via a learned connection matrix M :

$$r_j(t) = g \cdot \sum_{i=0}^{N-1} M_{ij} \cdot f_i(t) \quad (9)$$

where:

- M_{ij} is the connection weight from rhythm band i to neuron j
- $f_i(t)$ is the filter output of band i
- g is a global mapping gain parameter

3.4 Network Signal Flow

Rhythmogram Outputs [f, f, ..., f]

↓

Rhythmogram Connection Matrix M
(8 bands × N neurons)

↓

Rhythmic Inputs: $r = g \cdot M \cdot f$

↓

Neural Network Layer
Neuron 1 ↔ Neuron 2 ↔ ... ↔ Neuron N

Sample 1 Sample 2 Sample N
Each connection has weight w

↓

For each neuron j:

$a(t+1) = (.a(t) + w \cdot s(t) + r(t))$

If $a(t) > \text{Trigger}$ Sample j, Reset $a(t) = 0$

↓

Audio Output

4 Adaptive Learning Algorithm

4.1 Learning Objective

The system implements an online learning algorithm that adapts the rhythmogram-to-neuron connection matrix M to maximize the correlation between rhythmic events and neuron activations. The learning process is modulated by onset detection, emphasizing rhythmically salient moments.

4.2 Onset-Modulated Learning Rule

The update rule for connection weight M_{ij} is:

$$\Delta M_{ij} = -\eta \cdot \beta(t) \cdot \frac{\partial L}{\partial M_{ij}} - \delta \cdot M_{ij} \quad (10)$$

where:

- η is the learning rate
- $\beta(t)$ is the onset-dependent learning modulation

- L is the loss function
- δ is the weight decay parameter

The onset modulation factor $\beta(t)$ is computed as:

$$\beta(t) = 1 + \alpha \cdot \sum_{i:\text{onset at } t} O_i(t) \cdot M_{ij} \quad (11)$$

where:

- α is the onset bias parameter (0 = no onset influence, 1 = maximum onset influence)
- $O_i(t)$ is the onset strength for band i at time t

4.3 Loss Function

The system uses a prediction-based loss function comparing neuron activation to its assigned rhythmic band:

$$L_j = \frac{1}{2} (a_j(t) - f_{b(j)}(t))^2 \quad (12)$$

where $b(j)$ maps neuron j to its assigned frequency band.

The gradient with respect to the connection weight is:

$$\frac{\partial L_j}{\partial M_{ij}} = (a_j(t) - f_{b(j)}(t)) \cdot f_i(t) \quad (13)$$

4.4 Complete Learning Algorithm

5 Implementation Details

5.1 System Parameters

Table 1: Default System Parameters

Parameter	Symbol	Default Value	Range
Sample Rate	f_s	44100 Hz	—
Frame Size	—	512 samples	—
Filter Bands	N	8	1–16
Min Frequency	$f_{\min}$	0.125 Hz	—
Max Frequency	$f_{\max}$	16 Hz	—
Learning Rate	η	0.02	0.0–0.1
Weight Decay	δ	0.0005	0.0–0.01
Mapping Gain	g	0.2	0.0–1.0
Onset Bias	α	0.5	0.0–1.0
Onset Threshold	θ	1.0	0.0–10.0

5.2 Real-Time Performance

The system processes audio in real-time with latency determined by the frame size:

$$\text{Latency} = \frac{\text{Frame Size}}{f_s} = \frac{512}{44100} \approx 11.6 \text{ ms} \quad (14)$$

Algorithm 1 Onset-Modulated Rhythmogram Learning

Require: Audio stream, network configuration

Ensure: Adapted connection matrix M

```
1: Initialize  $M$  randomly or from preset
2: for each audio frame  $t$  do
3:   Extract rhythmogram features  $f(t)$ 
4:   Detect onsets  $O(t)$  across all bands
5:   for each neuron  $j$  do
6:     Compute rhythmic input:  $r_j(t) = g \cdot \sum_i M_{ij} \cdot f_i(t)$ 
7:     Update activation:  $a_j(t+1) = \phi_j(\lambda_j \cdot a_j(t) + \sum_k w_{kj} \cdot s_k(t) + r_j(t))$ 
8:     if  $a_j(t) \geq \theta_j$  then
9:       Trigger sample  $j$ 
10:       $a_j(t) \leftarrow 0$ 
11:    end if
12:    if learning enabled then
13:      Compute onset boost:  $\beta_j(t) = 1 + \alpha \cdot \sum_i O_i(t) \cdot M_{ij}$ 
14:      Compute error:  $e_j = a_j(t) - f_{b(j)}(t)$ 
15:      for each band  $i$  do
16:         $\Delta M_{ij} = -\eta \cdot \beta_j(t) \cdot e_j \cdot f_i(t) - \delta \cdot M_{ij}$ 
17:         $M_{ij} \leftarrow M_{ij} + \Delta M_{ij}$ 
18:         $M_{ij} \leftarrow \text{clip}(M_{ij}, -1, 1)$ 
19:      end for
20:    end if
21:  end for
22: end for
```

6 Musical Applications

6.1 Tempo-Relative Rhythm Analysis

The system can optionally adapt its filter bank to detected tempo. When BPM is known or estimated, the frequency bands shift to align with musically relevant subdivisions:

$$f'_i = f_i \cdot \frac{\text{BPM}}{120} \quad (15)$$

This ensures that filters remain locked to musical meter rather than absolute time.

6.2 Quantization

The system includes optional quantization that aligns neuron firing to a musical grid, supporting subdivisions from whole notes to 32nd note triplets. This bridges the continuous-time neural dynamics with discrete musical timing.

7 Complete System Diagram

AUDIO INPUT STREAM

↓

RHYTHMOGRAM ANALYZER

Envelope → Filter Bank → Onset
Extraction (8 bands) Detection

Outputs: $f(t) = [f(t), f(t), \dots, f(t)]$
 $o(t) = \{\text{onset events}\}$

↓

RHYTHMOGRAM-TO-NEURON MAPPING

$r(t) = g \cdot M \cdot f(t)$
where M is 8×N connection matrix

↓

NEURAL NETWORK LAYER

Neurons with recurrent connections
Update: $a(t+1) = (\cdot a(t) + w \cdot s(t) + r(t))$
Fire when $a(t)$, trigger sample, reset

↓

ADAPTIVE LEARNING MODULE

Onset-modulated weight updates
 $M = -\cdot(t) \cdot e \cdot f(t) - \cdot M$

↓

AUDIO OUTPUT
(Triggered Samples)

8 Experimental Results

8.1 Rhythmic Pattern Learning

In testing with various rhythmic inputs (4/4 drum loops, polyrhythmic patterns, irregular beats), the system demonstrates:

- Convergence of connection weights within 10–30 seconds
- Stable synchronization with input tempo
- Emergence of syncopated patterns when neurons interact

8.2 Onset Bias Effects

The onset bias parameter α significantly affects learning behavior:

- $\alpha = 0$: Learning responds to sustained energy, slower adaptation
- $\alpha = 0.5$: Balanced response to both continuous and transient features
- $\alpha = 1.0$: Strong emphasis on onsets, rapid adaptation to rhythmic attacks

9 Discussion

NeuronSeqSampler represents a synthesis of rhythm perception models [2] and neural network principles adapted from musical note modeling [1]. The key innovations include:

1. **Multi-scale rhythmic analysis:** Unlike fixed-tempo beat trackers, the logarithmic filter bank captures relationships across tempo ranges
2. **Onset-modulated learning:** The α parameter allows interpolation between energy-based and event-based learning
3. **Recurrent neural architecture:** Neuron interactions create emergent rhythmic patterns beyond direct rhythmogram mapping

The system enables real-time interaction where the network learns from and responds to rhythmic input while maintaining its own generative dynamics through recurrent connections.

10 Conclusion

We have presented the mathematical foundations and implementation of NeuronSeqSampler, a hybrid system combining rhythmic analysis and neural network principles for generative music sequencing. The architecture draws on established models of auditory rhythm perception while introducing novel onset-weighted learning mechanisms. Future work includes extending the tempo adaptation mechanisms and exploring hierarchical network structures for multi-timescale pattern generation.

References

- [1] Laine, U. K. (2000). *Analysis and Modeling of the Musical Note Onset*. Helsinki University of Technology, Department of Electrical and Communications Engineering. Available: <https://helda.helsinki.fi/server/api/core/bitstreams/6c68217b-752a-47bd-9b75-fb50e1fdc798/content>
- [2] Todd, N.P. (1994). The auditory “primal sketch”: A multiscale model of rhythmic grouping. *Journal of New Music Research*, 23(1), 25–70.

A Mathematical Notation Summary

Table 2: Mathematical Symbols and Definitions

Symbol	Description
N	Number of frequency bands
f_i	Center frequency of band i
$X_i(n)$	Filter output for band i at time n
$\Phi_i(n)$	Onset detection function for band i
$a_j(t)$	Activation of neuron j at time t
θ_j	Firing threshold for neuron j
λ_j	Leak rate (decay) for neuron j
w_{kj}	Connection weight from neuron k to j
M_{ij}	Rhythmogram connection weight from band i to neuron j
g	Global mapping gain
η	Learning rate
δ	Weight decay parameter
α	Onset bias (0=filter only, 1=onset only)
$\beta(t)$	Onset-dependent learning modulation
ϕ_j	Activation function for neuron j